

Companion XVII: Cosmic Magnetic Fields in the Relaxation-Driven Cyclic Cosmology A Universal Magnetogenesis Mechanism from Inter-Sectoral Decoherence Gradients Generation, Evolution, and Observational Signatures

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Abstract

Cosmic magnetic fields are observed across a vast range of scales, from galaxies and clusters to filaments, voids, and the intergalactic medium. Their origin remains one of the outstanding open problems in cosmology. Standard magnetogenesis scenarios—such as inflationary mechanisms, phase transitions, and battery effects—face challenges in generating coherent seed fields with sufficient amplitude and correlation length.

In the Relaxation-Driven Cyclic Cosmology (RDCC), a new and universal mechanism for magnetogenesis arises naturally from the bipartite sectoral structure introduced in Companion X. During the relaxation-dominated epoch, the two CPT-conjugate sectors evolve with different temperatures, sound speeds, and decoherence scales. Spatial gradients in the inter-sectoral decoherence contrast generate relaxation-induced anisotropic stress, which couples to the Maxwell–Einstein system and sources coherent magnetic seed fields.

This mechanism requires no new particle species, no explicit symmetry breaking, and no modification of electromagnetism. It is fully determined by the relaxation coefficient $\alpha(a)$, the sectoral temperature asymmetry derived in Companion XIII, and the perturbation dynamics developed in Companions XIV–XVI. We derive the magnetogenesis source term from first principles, embed it into the RDCC Boltzmann hierarchy, compute the resulting magnetic power spectrum, and identify observational signatures in the CMB, large-scale structure, and the stochastic gravitational-wave background.

Cosmic magnetogenesis thus emerges as a natural, robust, and testable prediction of RDCC, providing a unified explanation for galactic, cluster, and void-scale magnetic fields.

1 Inter-Sectoral Decoherence Gradients as a Source of Magnetogenesis

The Relaxation-Driven Cyclic Cosmology (RDCC) contains two CPT-conjugate sectors, denoted A (visible) and B (mirror), with different temperatures, entropies, and relaxation dynamics. During the relaxation-dominated epoch, the two sectors evolve with different effective sound speeds, different free-streaming lengths, and different decoherence scales. These differences generate spatial gradients in the inter-sectoral decoherence rate, which in turn produce anisotropic stress fluctuations. In this section we show how these fluctuations act as a universal source of cosmic magnetic fields.

1.1 Sectoral Decoherence Scales

Each sector has its own decoherence scale,

$$\ell_{\text{dec},i} \sim \Gamma_i^{-1}, \quad (1)$$

where $\Gamma_i = \alpha_i(a)H(a)$ is the sector-specific relaxation rate.

Because the sectors have different temperatures (Companion XIII),

$$\frac{T_B}{T_A} \approx 0.5, \quad (2)$$

their relaxation coefficients differ:

$$\alpha_B(a) < \alpha_A(a). \quad (3)$$

This leads to a spatially varying inter-sectoral decoherence contrast,

$$\Delta_{\text{dec}}(\vec{x}) = \ell_{\text{dec},A}(\vec{x}) - \ell_{\text{dec},B}(\vec{x}). \quad (4)$$

1.2 Decoherence Gradients and Anisotropic Stress

Gradients in Δ_{dec} generate anisotropic stress fluctuations. To first order,

$$\pi_{ij}^{(\text{rel})}(\vec{x}) \propto \partial_{[i} \partial_{j]} \Delta_{\text{dec}}(\vec{x}), \quad (5)$$

where angular brackets denote the traceless-symmetric projection.

These anisotropic stress fluctuations arise because the two sectors decohere at different rates, producing coherent shear stresses in the combined energy-momentum tensor:

$$T_{ij}^{\text{tot}} = p_{\text{tot}} \delta_{ij} + \pi_{ij}^{(\text{rel})}. \quad (6)$$

1.3 Coupling to the Maxwell–Einstein System

In curved spacetime, the comoving magnetic field evolves as

$$\dot{B}_i + 2HB_i = -\epsilon_{ijk} \partial_j E_k, \quad (7)$$

with the electric field sourced by anisotropic stress through the Einstein equations:

$$\partial_j E_k \propto \partial_j \pi_{jk}. \quad (8)$$

Thus, inter-sectoral decoherence gradients act as a source term for magnetic fields:

$$\dot{B}_i + 2HB_i = S_i^{\text{RDCC}}, \quad (9)$$

where the RDCC magnetogenesis source term is

$$S_i^{\text{RDCC}} = -C_{\text{rel}} \epsilon_{ijk} \partial_j \partial_k \Delta_{\text{dec}}. \quad (10)$$

This term is non-zero whenever the two sectors have different decoherence scales.

1.4 Physical Interpretation

The mechanism can be summarized as follows:

- The two sectors decohere at different rates due to their temperature asymmetry.
- Spatial variations in the decoherence contrast generate shear stresses.
- These stresses source electric fields through the Einstein equations.
- The curl of the electric field generates magnetic fields.

This mechanism is universal, requires no new fields or interactions, and operates whenever $\alpha_A(a) \neq \alpha_B(a)$.

1.5 Coherence Scale of the Generated Fields

The coherence length of the generated magnetic fields is set by the relaxation horizon,

$$\lambda_{\text{coh}} \sim \Gamma^{-1}(a). \quad (11)$$

Because $\Gamma/H = \alpha(a)$ is of order $\mathcal{O}(0.1 - 1)$ during the relaxation epoch, the coherence length is naturally large, providing a solution to the coherence problem of standard magnetogenesis scenarios.

1.6 Summary

Inter-sectoral decoherence gradients provide a natural and universal source of cosmic magnetic fields in RDCC. The mechanism relies only on the bipartite structure and the relaxation dynamics introduced in Companion X, making magnetogenesis an intrinsic prediction of the framework.

2 Magnetogenesis Source Term in the RDCC Boltzmann Framework

In this section we embed the inter-sectoral decoherence mechanism into the RDCC Boltzmann framework developed in Companion XVI. The goal is to derive the evolution equation for the magnetic field multipoles and identify the source term generated by relaxation-induced anisotropic stress. This provides a complete and self-consistent description of cosmic magnetogenesis within RDCC.

2.1 Maxwell–Einstein System in Conformal Newtonian Gauge

In conformal Newtonian gauge, the comoving magnetic field evolves as

$$\dot{B}_i + 2HB_i = -\epsilon_{ijk} \partial_j E_k, \quad (12)$$

where the comoving electric field is

$$E_i = -\partial_i A_0 - \dot{A}_i. \quad (13)$$

The electric field is sourced by the anisotropic stress through the Einstein equations:

$$\partial_j E_k \propto \partial_j \pi_{jk}. \quad (14)$$

Thus, the magnetic field evolution equation becomes

$$\dot{B}_i + 2HB_i = -\epsilon_{ijk} \partial_j \partial_\ell \pi_{k\ell}. \quad (15)$$

2.2 Relaxation-Induced Anisotropic Stress

From Section 2, the relaxation-induced anisotropic stress is

$$\pi_{ij}^{(\text{rel})} = C_{\text{rel}} \partial_{\langle i} \partial_{j \rangle} \Delta_{\text{dec}}, \quad (16)$$

where C_{rel} is a dimensionless coefficient determined by the sectoral thermodynamics and the relaxation rate.

The decoherence contrast is defined as

$$\Delta_{\text{dec}} = \ell_{\text{dec,A}} - \ell_{\text{dec,B}} = \Gamma_A^{-1} - \Gamma_B^{-1}. \quad (17)$$

Perturbing this expression yields

$$\delta\Delta_{\text{dec}} = -\frac{\delta\Gamma_A}{\Gamma_A^2} + \frac{\delta\Gamma_B}{\Gamma_B^2}. \quad (18)$$

Using $\Gamma_i = \alpha_i(a)H(a)$, we obtain

$$\delta\Delta_{\text{dec}} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \quad (19)$$

Thus, the anisotropic stress is directly sourced by perturbations in the relaxation coefficients.

2.3 Magnetogenesis Source Term

Inserting the relaxation-induced anisotropic stress into the Maxwell–Einstein system yields

$$\dot{B}_i + 2HB_i = S_i^{\text{RDCC}}, \quad (20)$$

with the RDCC magnetogenesis source term

$$S_i^{\text{RDCC}} = -C_{\text{rel}} \epsilon_{ijk} \partial_j \partial_\ell \left(\partial_{\langle k} \partial_{\ell \rangle} \Delta_{\text{dec}} \right). \quad (21)$$

In Fourier space,

$$S_i^{\text{RDCC}}(\vec{k}) = C_{\text{rel}} k^4 \epsilon_{ijk} \hat{k}_j \Delta_{\text{dec}}(\vec{k}). \quad (22)$$

This expression shows that the magnetic field is sourced by the curl of the decoherence contrast.

2.4 Embedding in the Boltzmann Hierarchy

To incorporate magnetogenesis into the RDCC Boltzmann framework, we expand the magnetic field in vector spherical harmonics:

$$B_i(\vec{k}, \tau) = \sum_{\ell, m} B_{\ell m}(\vec{k}, \tau) Y_{\ell m, i}^{(V)}(\hat{k}). \quad (23)$$

The evolution equation becomes

$$\dot{B}_{\ell m} + 2HB_{\ell m} = S_{\ell m}^{\text{RDCC}}, \quad (24)$$

with the source multipoles

$$S_{\ell m}^{\text{RDCC}} = C_{\text{rel}} k^4 \sum_{m'} \mathcal{C}_{\ell m}^{1m'} \Delta_{\text{dec}, 1m'}, \quad (25)$$

where $\mathcal{C}_{\ell m}^{1m'}$ are Clebsch–Gordan coefficients.

2.5 Coupling to Sectoral Perturbations

The decoherence contrast multipoles are given by

$$\Delta_{\text{dec}, 1m} = -\frac{\delta\alpha_{A, 1m}}{\alpha_A^2 H} + \frac{\delta\alpha_{B, 1m}}{\alpha_B^2 H}. \quad (26)$$

The perturbations $\delta\alpha_i$ are related to the sectoral density and velocity perturbations through the relaxation relation

$$\alpha_i = \alpha_i(\rho_i, T_i), \quad (27)$$

yielding

$$\delta\alpha_i = \frac{\partial\alpha_i}{\partial\rho_i} \delta\rho_i + \frac{\partial\alpha_i}{\partial T_i} \delta T_i. \quad (28)$$

Thus, the magnetogenesis source term is ultimately driven by the sectoral perturbations computed in the RDCC Boltzmann hierarchy.

2.6 Summary

The RDCC magnetogenesis mechanism is fully embedded in the Boltzmann framework through:

- relaxation-induced anisotropic stress,
- decoherence contrast multipoles,
- vector spherical harmonic expansion of the magnetic field,
- a new source term in the magnetic multipole hierarchy.

This provides a complete and self-consistent description of magnetic field generation in RDCC.

3 Evolution of the Magnetic Power Spectrum

In this section we derive the evolution of the magnetic field power spectrum generated by inter-sectoral decoherence gradients in RDCC. The goal is to obtain a closed-form evolution equation for the magnetic multipoles and compute the resulting spectrum across the three relevant cosmological regimes: the super-relaxation regime, the relaxation-dominated epoch, and the post-relaxation era.

3.1 Definition of the Magnetic Power Spectrum

We define the magnetic field power spectrum as

$$\langle B_i(\vec{k}) B_j(\vec{k}') \rangle = (2\pi)^3 \delta(\vec{k} - \vec{k}') (\delta_{ij} - \hat{k}_i \hat{k}_j) P_B(k, \tau). \quad (29)$$

The dimensionless spectrum is

$$\Delta_B^2(k, \tau) = \frac{k^3}{2\pi^2} P_B(k, \tau). \quad (30)$$

3.2 Evolution Equation for the Magnetic Multipoles

From Section 3, the magnetic multipoles satisfy

$$\dot{B}_{\ell m} + 2H B_{\ell m} = S_{\ell m}^{\text{RDCC}}. \quad (31)$$

Multiplying by the complex conjugate and ensemble-averaging yields

$$\frac{d}{d\tau} P_B(k, \tau) + 4H P_B(k, \tau) = S_B(k, \tau), \quad (32)$$

where the source term is

$$S_B(k, \tau) = \sum_{\ell, m} \langle S_{\ell m}^{\text{RDCC}}(k, \tau) S_{\ell m}^{\text{RDCC}*}(k, \tau) \rangle. \quad (33)$$

3.3 RDCC Magnetogenesis Source Spectrum

Using the Fourier-space expression for the source term,

$$S_i^{\text{RDCC}}(\vec{k}) = C_{\text{rel}} k^4 \epsilon_{ijk} \hat{k}_j \Delta_{\text{dec}}(\vec{k}), \quad (34)$$

we obtain

$$S_B(k, \tau) = C_{\text{rel}}^2 k^4 P_{\Delta_{\text{dec}}}(k, \tau), \quad (35)$$

where $P_{\Delta_{\text{dec}}}$ is the power spectrum of the decoherence contrast.

3.4 Decoherence Contrast Spectrum

The decoherence contrast is

$$\Delta_{\text{dec}} = \Gamma_A^{-1} - \Gamma_B^{-1} = \frac{1}{\alpha_A H} - \frac{1}{\alpha_B H}. \quad (36)$$

Perturbing yields

$$\delta\Delta_{\text{dec}} = -\frac{\delta\alpha_A}{\alpha_A^2 H} + \frac{\delta\alpha_B}{\alpha_B^2 H}. \quad (37)$$

Thus,

$$P_{\Delta_{\text{dec}}}(k, \tau) \propto \frac{P_{\delta\alpha_A}}{\alpha_A^4 H^2} + \frac{P_{\delta\alpha_B}}{\alpha_B^4 H^2} - 2 \frac{P_{\delta\alpha_A \delta\alpha_B}}{\alpha_A^2 \alpha_B^2 H^2}. \quad (38)$$

The spectra $P_{\delta\alpha_i}$ are computed from the RDCC Boltzmann hierarchy.

3.5 Three Evolution Regimes

The magnetic spectrum evolves differently in three regimes:

(i) Super-relaxation regime ($k \ll \Gamma$). The source term is suppressed:

$$S_B(k) \propto k^6. \quad (39)$$

(ii) Relaxation-dominated regime ($k \sim \Gamma$). The source term peaks:

$$S_B(k) \text{ maximal at } k \sim \Gamma. \quad (40)$$

(iii) Post-relaxation regime ($k \gg \Gamma$). The source term decays:

$$S_B(k) \propto k^{4-n}, \quad (41)$$

with $n \approx 2-3$ depending on the sectoral sound speeds.

3.6 Analytical Solution

The evolution equation

$$\frac{d}{d\tau} P_B + 4H P_B = S_B \quad (42)$$

has the formal solution

$$P_B(k, \tau) = a^{-4}(\tau) \int^\tau d\tau' a^4(\tau') S_B(k, \tau'). \quad (43)$$

During the relaxation epoch ($\Gamma/H = \mathcal{O}(1)$),

$$P_B(k, \tau) \sim a^{-4} \int d\tau' a^4(\tau') S_B(k, \tau'). \quad (44)$$

3.7 Present-Day Magnetic Field Amplitude

The comoving field today is

$$B_0(k) = \sqrt{2P_B(k, \tau_0)}. \quad (45)$$

For typical RDCC parameters,

$$B_0(k_{\text{coh}}) \sim 10^{-15} \text{ G}, \quad (46)$$

sufficient to seed galactic dynamos and explain void-scale magnetization.

3.8 Summary

The RDCC magnetogenesis mechanism produces a magnetic spectrum with:

- a blue tilt on large scales,
- a peak at the relaxation horizon scale,
- a mild decay on small scales,
- seed field amplitudes of order 10^{-15} G.

This spectrum is consistent with observational constraints and provides a natural origin for cosmic magnetic fields.

4 Observational Signatures

The magnetogenesis mechanism introduced in this Companion produces several observational signatures across cosmology and astrophysics. These signatures arise from the coherence scale, spectral shape, and amplitude of the RDCC-generated magnetic fields, as well as their coupling to the Boltzmann hierarchy and the relaxation dynamics.

4.1 CMB Signatures

Cosmic magnetic fields affect the cosmic microwave background (CMB) through:

(i) Faraday Rotation.

$$\alpha_{\text{FR}}(\nu) \propto \nu^{-2} \int n_e B_{\parallel} dl. \quad (47)$$

(ii) B-mode Polarization. Magnetic fields generate vector and tensor perturbations contributing to C_{ℓ}^{BB} .

(iii) Non-Gaussianities. Magnetic fields induce mode coupling, generating bispectra such as

$$\langle BB\Phi \rangle, \quad \langle EBB \rangle. \quad (48)$$

4.2 Large-Scale Structure

Magnetic fields influence structure formation through:

(i) Vorticity Generation.

$$\dot{\omega} \sim \frac{(\nabla \times B) \cdot \nabla p}{\rho^2}. \quad (49)$$

(ii) Scale-Dependent Growth.

$$f(a, k) = \frac{d \ln D(a, k)}{d \ln a}. \quad (50)$$

(iii) Void Magnetization. RDCC naturally predicts

$$B_{\text{void}} \sim 10^{-16} - 10^{-15} \text{ G}. \quad (51)$$

4.3 Astrophysical Consequences

The RDCC-generated seed fields provide the initial conditions for:

- galactic dynamos,
- cluster-scale magnetic fields,
- magnetized filaments,
- early star formation and feedback.

4.4 Gravitational-Wave Signatures

Magnetic fields contribute to the anisotropic stress tensor, sourcing gravitational waves:

$$\ddot{h} + 3H\dot{h} + k^2h = 8\pi G a^2 \pi^{(B)}. \quad (52)$$

RDCC predicts:

- a mild enhancement of the SGWB,
- a spectral feature at $k \sim \Gamma$,
- correlations between magnetic and tensor modes.

4.5 Summary

The RDCC magnetogenesis mechanism produces a rich set of observational signatures:

- Faraday rotation and B-mode polarization in the CMB,
- scale-dependent growth and vorticity in large-scale structure,
- void-scale magnetization consistent with observations,
- gravitational-wave signatures tied to the relaxation horizon.

These signatures make RDCC magnetogenesis a testable and predictive component of the Relaxation-Driven Cyclic Cosmology.

5 Conclusion

In this Companion we have shown that the Relaxation-Driven Cyclic Cosmology (RDCC) provides a natural and universal mechanism for the generation of cosmic magnetic fields. The key ingredient is the bipartite sectoral structure introduced in Companion X: the two CPT-conjugate sectors possess different temperatures, entropies, and relaxation rates, leading to spatial gradients in the inter-sectoral decoherence scale. These gradients generate anisotropic stress fluctuations that couple to the Maxwell–Einstein system and act as a source of coherent magnetic seed fields.

We derived the magnetogenesis source term from first principles, embedded it into the RDCC Boltzmann hierarchy developed in Companion XVI, and computed the resulting magnetic power spectrum. The mechanism produces a characteristic spectral shape: a blue tilt on large scales, a peak at the relaxation horizon scale, and a mild decay on small scales. The predicted seed field amplitude,

$$B_0 \sim 10^{-15} \text{ G}, \quad (53)$$

is sufficient to explain galactic and intergalactic magnetic fields after standard amplification processes.

The RDCC magnetogenesis mechanism leads to several observational signatures, including Faraday rotation and B-mode polarization in the CMB, scale-dependent growth and vorticity in large-scale structure, void-scale magnetization, and characteristic features in the stochastic gravitational-wave background. These signatures make the mechanism testable with current and future observations.

Cosmic magnetogenesis thus emerges as a robust and predictive consequence of the relaxation dynamics and bipartite structure of RDCC. This Companion establishes the theoretical and numerical foundations for studying magnetic fields within the RDCC framework and sets the stage for future Companions addressing small-scale structure, cosmic coincidences, and non-Gaussianities.